Latest Instructions for Milestone 5 Simplification:

Let it be known that there will be no graph database in our project. Anything from the rubric that specifically mentions a graph database will be replaced with a rubric item instructing reviewers to assign full credit.

MILESTONE 5: Multi-Model Integration & Hybrid RAG Retrieval Engine

Deadline: Sunday, November 1, 11:59 PM
Peer Review Due: Wednesday, November 4, 11:59 PM

Objectives

Integrate IoT telemetry (document store) ~~and supply chain graphs~~ into a unified retrieval system. Build an end-to-end RAG pipeline that answers natural language questions using context from multiple data sources.

What You'll Do

1. **Ingest IoT telemetry into document store**:
    - Load JSON telemetry logs into MongoDB (or similar document database)
    - Index by timestamp, equipment_id, and error codes
    - Write queries to retrieve relevant diagnostic data based on battery failures

~~2. **Model supply chain in graph database**:~~
    - ~~Load supplier network into Neo4j (or similar graph database)~~
    - ~~Create nodes for suppliers, components, and assembly plants~~
    - ~~Define relationships representing dependencies and single-source risks~~
    - ~~Write Cypher queries (or equivalent) to traverse supplier chains~~

2. **Build hybrid retrieval function**:
    - Combine SQL filters (equipment metadata, warranty status)
    - Vector similarity search (technical documentation embeddings)
    - Document queries (IoT telemetry patterns)
    - ~~Graph traversal (supply chain context)~~
    - Create a unified retrieval pipeline that gathers context from all four data sources

2. **Implement RAG generation pipeline**:

    - Assemble retrieved chunks into structured prompt template
    - Pass context to LLM (OpenAI GPT-4, Anthropic Claude, or local models via Ollama)
    - Generate grounded answers with source citations
    - Track provenance: ensure every fact in the answer traces to a specific data source
2. **Test with realistic queries**:
    - "What are common failure modes for Model X-7000 batteries in high-temperature environments?"
    - ~~"Which suppliers provide lithium cells for our commercial energy storage units?"~~
    - "Show me service history for equipment unit SN-4829 and related warranty claims"

Deliverables

- **Code Artifacts**:
    - Python hybrid retrieval script integrating all data sources
    - End-to-end RAG pipeline script
    - Configuration files for document and graph databases
    - Sample prompt templates
    - Test query results with source provenance mapping
- **Video Screencast** (3-5 minutes):
    - Demonstrate live natural language query submission
    - Show retrieval process gathering context from multiple databases
    - Display LLM-generated answer with source citations
    - Explain how you ensured accuracy and prevented hallucination

Success Criteria

- All ~~four~~ three data sources successfully integrated
- Hybrid retrieval returns relevant context from multiple databases
- RAG pipeline generates accurate, grounded answers

- Every statement in generated answers can be traced to source data
- Sub-5-second end-to-end query latency (from question to answer)

Beyond RAG: Other Vector Database Applications

While this course focuses on RAG for the case competition, vector databases power many other AI applications:

- **Semantic search engines**: Finding conceptually similar documents regardless of exact keyword matches
- **Recommendation systems**: "Customers who liked X also liked Y" based on embedding similarity
- **Anomaly detection**: Identifying outlier patterns in IoT telemetry or transaction data
- **Duplicate detection**: Finding near-duplicate records in messy datasets
- **Image and audio search**: Retrieving visually or acoustically similar media

Understanding vector databases prepares you for the full spectrum of modern AI-powered data applications.

Milestone 5 Rubric:

Reviewer Instructions: Evaluate your peer's submission as a Technical Manager reviewing a peer's work. Provide specific, constructive feedback that will help them improve for the next milestone. Provide comments in the document as well as in the rubric. Reference specific artifacts (SQL scripts, ERD, video timestamps) in your comments.

Criterion	Exemplary (4 pts)	Proficient (3 pts)	Developing (2 pts)	Needs Improvement (1 pt)	Not Evident (0 pts)
1. Document Store Implementation (IoT Telemetry)	Excellent document database deployment (MongoDB or similar) with IoT telemetry properly indexed by timestamp, equipment_id, and error codes. Efficient queries retrieve relevant diagnostic data for battery failures. Well-designed schema with appropriate nested document structure.	Good document store implementation with JSON telemetry loaded and indexed. Most queries functional with adequate indexing strategy. Schema reasonably designed for telemetry use cases.	Basic document database setup with telemetry data loaded but limited indexing. Queries functional but inefficient or missing key indexes. Schema design has gaps in organization.	Minimal document store functionality. Poor indexing or inappropriate schema design. Queries extremely slow or return incomplete results.	No document database implemented or completely non-functional.

2. ~~Graph Database Implementation (Supply Chain)~~ **Assign Full Credit**	~~Exceptional graph database deployment (Neo4j or similar) with complete supplier network modeled. Nodes and relationships clearly defined with appropriate properties. Cypher queries (or equivalent) effectively traverse supplier chains, identify dependencies, and analyze single-source risks.~~	~~Good graph database with supplier network loaded. Most nodes and relationships properly defined. Graph traversal queries functional demonstrating basic supply chain analysis capabilities.~~	~~Basic graph database setup with some supplier data loaded but incomplete relationship modeling. Limited query complexity or unclear graph structure. Missing key supplier tiers or dependencies.~~	~~Minimal graph functionality. Poor data modeling or relationships incorrectly defined. Queries don't effectively demonstrate graph traversal or supply chain analysis.~~	~~No graph database implemented or completely non-functional.~~
3. Hybrid Retrieval Function Design	Outstanding hybrid retrieval implementation seamlessly combining SQL metadata filters, vector similarity	Good hybrid retrieval function integrating most data sources (SQL, vector, document,	Basic hybrid retrieval with some data sources integrated but incomplete coverage or poor	Minimal hybrid functionality. Significant gaps in data source integration or poor coordination. Retrieval heavily biased toward	No hybrid retrieval function or function fails to integrate multiple data sources.

	search, document queries, ~~and graph traversal~~. Well-architected code with modular design allowing each data source to contribute relevant context. Intelligent query routing based on question type.	graph). Functional context gathering with reasonable query coordination. Minor gaps in integration or optimization.	coordination between sources. Limited sophistication in combining results from multiple databases.	one source while underutilizing others.	
4. End-to-End RAG Pipeline Quality	Exceptional RAG pipeline with sophisticated prompt template assembly, clean context formatting, and effective LLM integration (OpenAI, Anthropic, or local models). Generates accurate,	Good RAG pipeline functional with reasonable prompt engineering. LLM generates relevant answers using retrieved context. Adequate handling of typical queries with minor	Basic RAG pipeline operational but prompt templates simplistic or context assembly inefficient. LLM responses sometimes miss relevant context or produce verbose/unfocused answers.	Minimal RAG functionality. Poor prompt engineering resulting in low-quality answers. LLM frequently ignores provided context or produces irrelevant responses. Significant reliability issues.	No RAG pipeline implemented or pipeline completely non-functional.

	well-structured answers that directly address questions using retrieved context. Excellent error handling and edge case management.	issues in edge cases or formatting.	Limited error handling.		
5. Source Provenance & Citation Tracking	Exceptional provenance system ensuring 100% traceability of every fact in generated answers to specific source records. Citations include database type, record ID, document name, and timestamp. Clear mapping between answer statements and source data	Good provenance tracking with most facts traceable to sources. Citations include essential identifiers. Some minor gaps in traceability or citation formatting but core provenance maintained.	Basic provenance implementation with partial traceability. Some facts cited but others lack clear source attribution. Citation format inconsistent or incomplete identifiers provided.	Minimal provenance effort. Significant gaps in source traceability. Many answer statements cannot be verified against source data. Poor or missing citation system.	No source provenance tracking or citations completely absent from generated answers.

	demonstrable in video.				
6. Performance & Latency (Sub-5-Second Target)	Outstanding end-to-end performance with median query latency well under 5 seconds from question submission to complete answer generation. Includes latency profiling breaking down retrieval vs. generation time. Professional benchmarking with statistical analysis (P50, P95, P99).	Good performance meeting or approaching sub-5-second target. Adequate benchmarking demonstrating reasonable latency. Some profiling of pipeline stages with identification of major time contributors.	Basic performance testing showing latency but exceeds 5-second target significantly (5-10 seconds). Limited profiling or unclear bottleneck identification. Performance inconsistent across queries.	Poor performance with latency regularly exceeding 10 seconds. Minimal benchmarking effort or unreliable measurements. No meaningful performance analysis or optimization attempts.	No performance benchmarking provided or system too slow/unreliable to measure meaningfully.
7. Video Demonstration Quality	Clear, professional 3-5 minute screencast	Good video showing most components: query	Video present but incomplete demonstration. Some data	Poor video quality or missing major components. Difficult to verify hybrid retrieval or	No video provided or video doesn't demonstrate

	demonstrating live natural language query submission, hybrid retrieval across all four data sources, context assembly, LLM answer generation with source citations, and provenance verification. Excellent narration explaining system architecture and data flow. Within time limit.	submission, retrieval from multiple sources, and answer generation with citations. Narration mostly clear explaining key components. Within time limit.	sources or RAG components not clearly shown. Narration unclear or rushed. Significantly over/under time limit. Provenance not clearly demonstrated.	RAG functionality. Significantly over/under time limit. Minimal explanation of approach or integration.	working RAG system with multi-model integration.
8. Technical Understanding & Integration Rationale	Exceptional explanation of multi-model architecture decisions, hybrid retrieval strategy, RAG prompt	Good explanation of major architectural decisions and integration approach. Adequate	Basic explanation with limited depth. Some understanding of different database models but gaps in explaining	Minimal explanation of decisions. Choices appear arbitrary or poorly justified. Limited understanding of multi-model architecture	No explanation of design decisions or demonstrates fundamental misunderstanding of multi-model

	engineering, and LLM selection. Demonstrates deep understanding of polyglot persistence, when to use each database type, and trade-offs in RAG system design. Clear articulation of how different data sources complement each other.	discussion of database choices and RAG pipeline design. Basic understanding of multi-model systems and retrieval strategies demonstrated.	integration rationale or RAG design choices. Minimal discussion of trade-offs.	or RAG system design principles.	integration or RAG concepts.